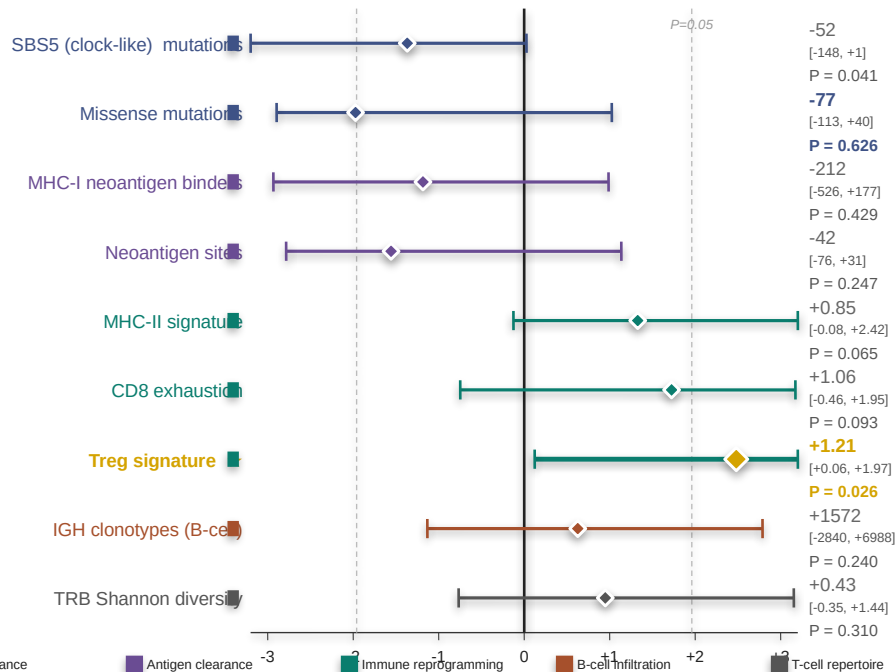


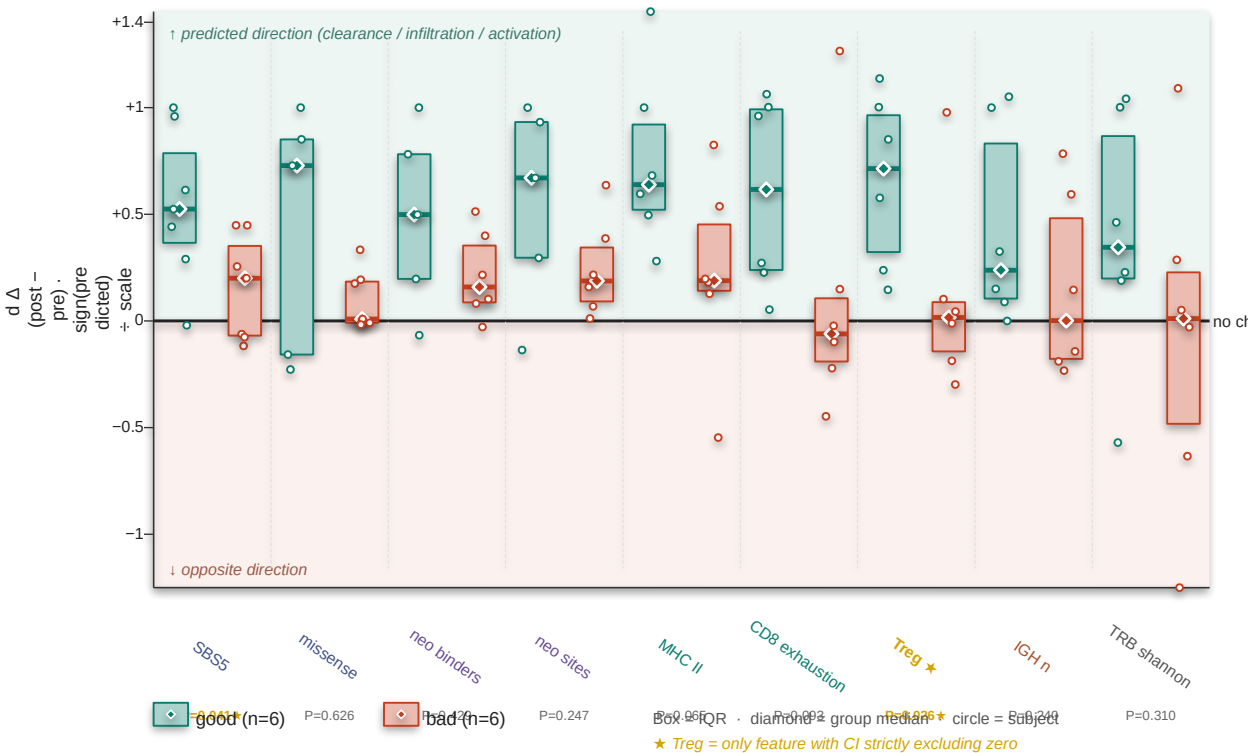
A



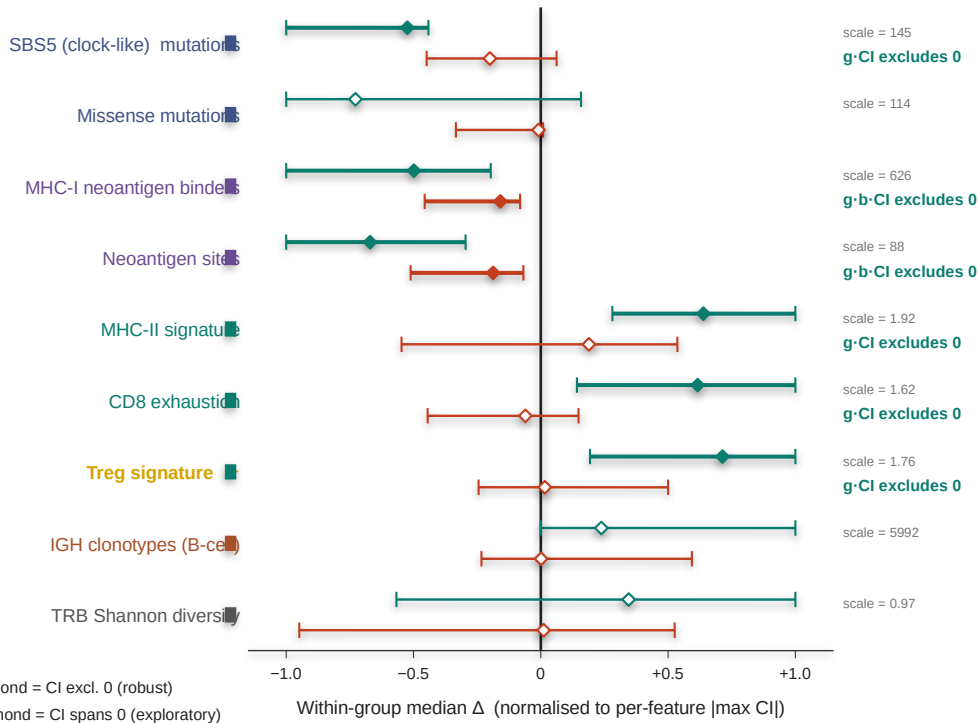
★ Only Treg Δ has a between-group BCa CI strictly excluding zero (robust $\alpha = 12$).

Other cascade features are framed as exploratory phenomenology; see Panel C for within-group robustness.

B

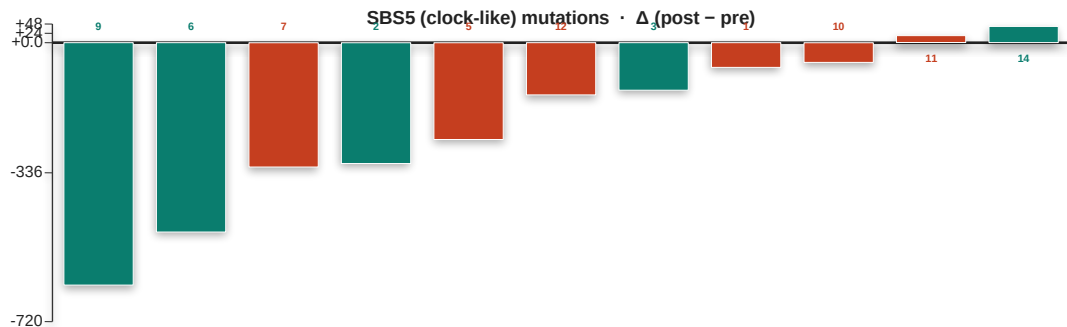
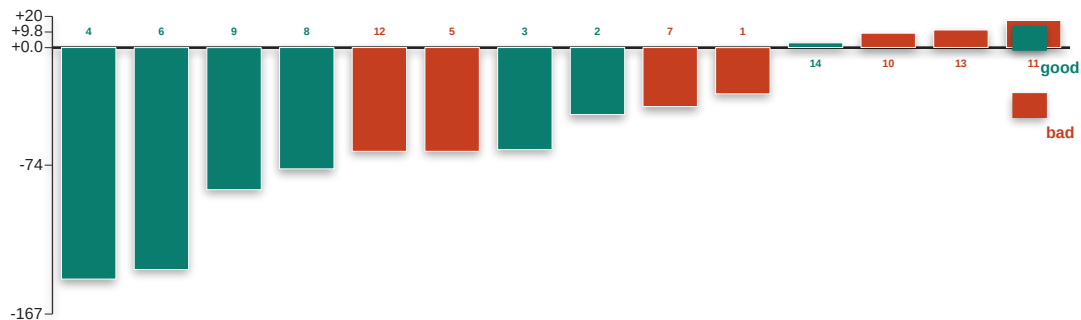


C



Within-group CIs excluding zero: SBS5, neo_binders, neo_sites, MHC-II, CD8 exhaustion, Treg. Consistent cascade within responders; between-group only Treg.

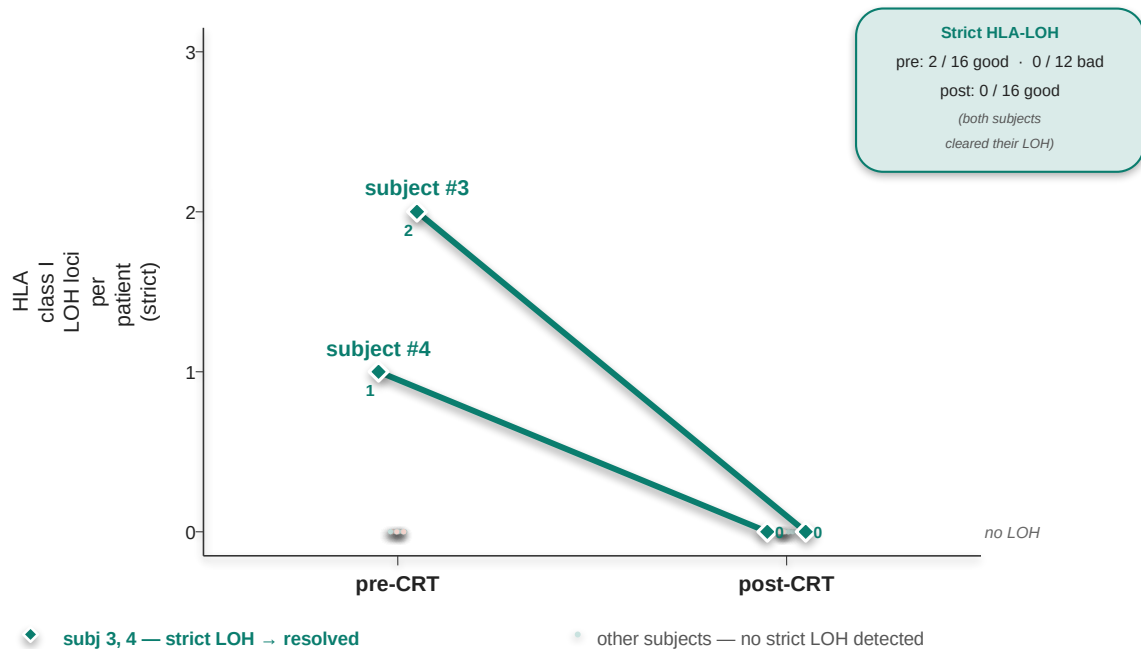
D



MHC-I neoantigen binders · Δ (post – pre)

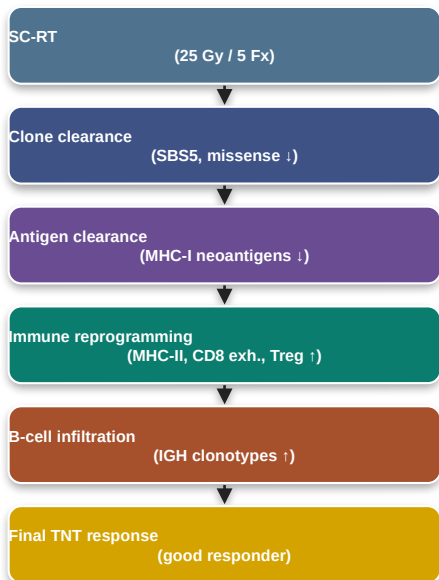
Per-subject Δ sorted ascending. Subjects #s shown near bar bases. Within good: SBS5 Δ median -76 (CI [-145, -64]), neo_binders Δ median -312 (CI [-626, -123]). Within-good CIs exclude zero; between-group CIs span zero → exploratory.

E



Bonferroni-corrected IMGT allele-count criterion ($|\Delta\text{ratio}| \geq 0.20$, Fisher $P < 0.01$). Fisher $P = 0.49$ between-group; anecdotal.

F



★ Treg CI
robust

Baseline → cascade Δ ?

NO

0 / 36 pairs $P < 0.05$

DSB → CD8-cyt $r = -0.07$

Static $\perp$ Dynamic

(orthogonal, not cascading)

Cascade is phenomenology, not a causal downstream of the baseline predictor.